How much can a fractional audit overstate coalition savings? Exact bounds for pooled newsvendor costs under a standalone-cost floor

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Abstract

Producers that pool their imbalance risk, such as wind farms bidding jointly, share the expected cost of the pooled offer and must check that no group of them would do better on its own. Checking every coalition is hard, so the check is often relaxed to fractional memberships. We ask how far that relaxation can overstate the largest saving available to an actual coalition when every producer is known to pay at least a fraction r of its standalone cost and k producers are excluded from the search. For equal shares we give the exact worst case for every number of producers, every depth k and every r: it equals the elementary ceiling $(1-r)(n-k)/n$ on a union of explicit intervals and is strictly smaller elsewhere, with closed forms in all regimes. For arbitrary shares we give the exact worst case at the deepest nonterminal level and four families of explicit worst-case laws at every depth, whose best value matched a certified upper bound in all 1068 random instances computed. Every bound is attained by a finite demand law. The constructions rest on the fact that costs of symmetric demand laws are zonotope support functions; the upper bounds rest on triangle-inequality certificates. With asymmetric imbalance penalties the ceiling remains valid and every worst case persists; we give the exact worst case at the deepest level for every penalty ratio, and for equal shares show that the windows close once a shortfall costs at least $2(m-1)/m$ times as much as a surplus.

1 Introduction

Renewable producers that offer their output jointly pay imbalance penalties on the pooled position rather than on their own. Pooling lowers the expected penalty, and the saving must be shared. A cost split is acceptable only if no group of producers would pay less by leaving and offering on its own: the split must lie in the core of the pooling game (Baeyens et al., 2013; Chakraborty et al., 2018). For newsvendor-type games the core is nonempty (Hartman et al., 2000; Müller et al., 2002; Montrucchio et al., 2012), but deciding whether a given split lies in it is NP-hard (Chen and Zhang, 2009).

A natural tractable check replaces actual coalitions by fractional ones: a producer may join with a fraction of its output, and the largest saving over fractional memberships is a convex program. The fixed-allocation version of this relaxation appears in Burer and Dror (2005, Section 5); fractional and partially continuous memberships have a long history in cooperative game theory (Aubin, 1981; Samet and Zemel, 1984; Denault, 2001; Bonnans and André, 2008), and integrality gaps of relaxations are closely tied to approximate cores in combinatorial games (Deng et al., 1999; Goemans and Skutella, 2004; Caprara and Letchford, 2010). The relaxation can only overstate the saving. The question of this paper is by how much.

Without further information the answer is discouraging: the overstatement can be as large as the charges of the producers kept in the search (Proposition 1). But an auditor usually knows more. We assume that every producer pays at least a fraction r of its standalone cost, and ask for

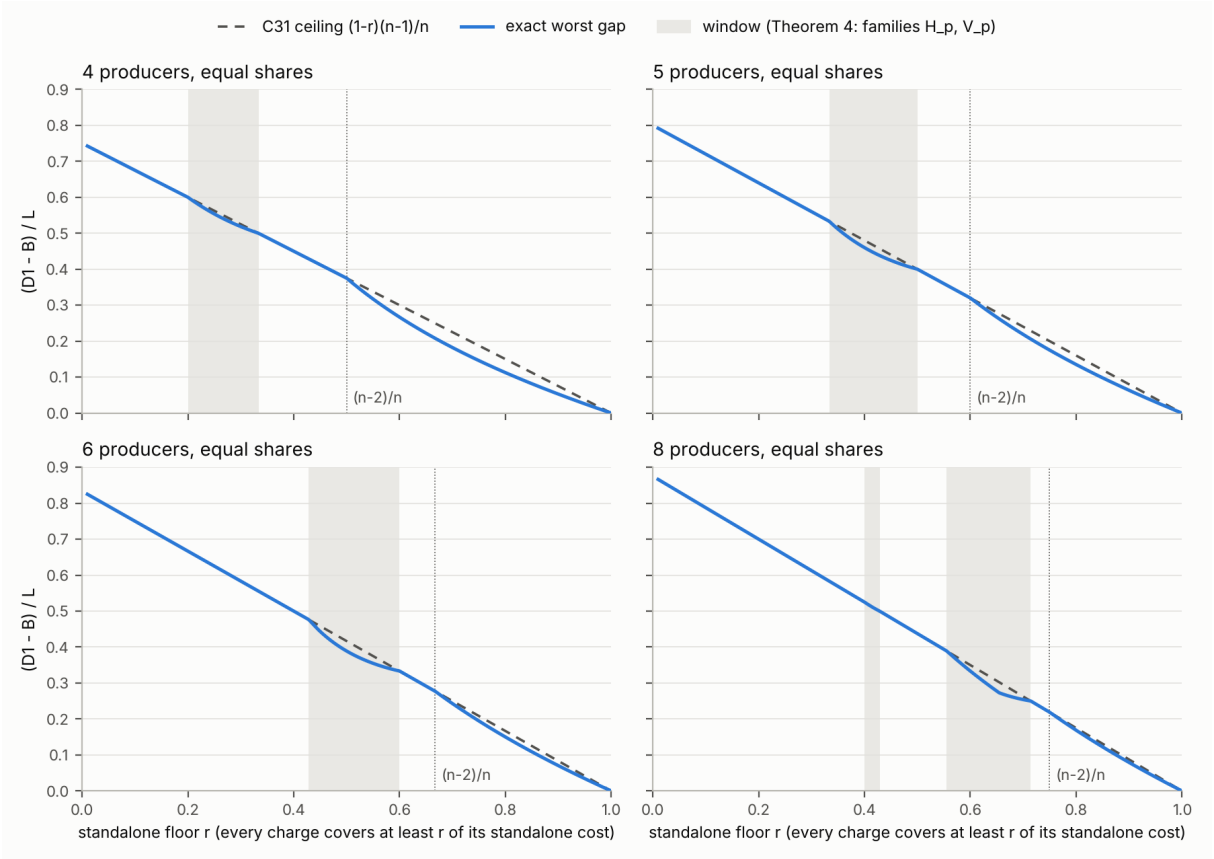


Figure 1: Exact worst-case overstatement $(D_1 - B)/L$ at depth one for equal shares, against the standalone floor r, with the elementary ceiling $(1 - r)(n - 1)/n$ (dashed). Shaded bands are the windows of Theorem 5; the dotted line marks $(n - 2)/n$.

the exact worst case as a function of r, of the shares and of the number k of producers excluded from the search.

Results. For equal shares the answer is complete (Theorems 4 and 5, Figure 1). The elementary ceiling $(1 - r)(n - k)/n$ is exact on a union of explicit intervals of r; above $(m - 1)/(m + 1)$, with $m = n - k$, the worst case is $(m - 1)(1 - r^2)/(2nr)$; and in the remaining "windows" it is the larger of two explicit families, with a quadratic in r deciding which. For arbitrary shares the worst case at depth $n - 2$ has three regimes per pair of producers (Theorem 7). At every depth, four explicit families give lower bounds (Theorem 8) whose maximum matched a certified upper bound in every one of 1068 random instances, which we state as a conjectured closed form (Conjecture 9). With generic shares the elementary ceiling turned out to be exact whenever $r \leq 0.4$ in our samples, and never when $r \geq 0.7$.

Method. Every lower bound is attained by an explicit finite, centrally symmetric demand law. Such laws have costs $c(x) = \sum_k p_k |x_k^\top x|$, support functions of zonotopes, a classical identity (Koshevoy and Mosler, 1997; Molchanov et al., 2012); the constructions are lists of generators. Every upper bound is a nonnegative combination of triangle-inequality certificates. Requiring positive definite covariance and strict stability of every coalition does not change any of the suprema (Proposition 10).

2 Model

Let $N = \{1, \dots, n\}$, $n \geq 3$, and let $Y = (Y_1, \dots, Y_n)$ be bounded, nonnegative outputs with arbitrary dependence. Imbalance is priced by the symmetric absolute loss $\ell(q, t) = |t - q|$. For a membership vector $z \in \mathbb{R}^n_+$ the *physical fractional cost* is

$$c(z) = \min_{q \in \mathbb{R}} \mathbb{E}|z^\top Y - q|.$$

It is nonnegative, convex and positively homogeneous; with signed arguments it is a seminorm. A frozen grand offer q_0 has expected cost $L = \mathbb{E}|\mathbf{1}^\top Y - q_0| \geq c(\mathbf{1})$, and producer j is charged $a_j = Lw_j$ for positive shares w summing to one. We normalize $L = 1$ throughout, so $a = w$. The excess of a membership is $f(z) = a^\top z - c(z)$, the largest positive saving of an actual proper coalition is $B = \max\{0, \max_{\emptyset \neq S \subsetneq N} f(\mathbf{1}_S)\}$, and at depth k the audit computes

$$D_k = \max\Big\{0,\ \max_{|N \setminus S| < k} f(\mathbf{1}_S),\ \max_{|J|=k}\ \max_{z \in [0,1]^n,\ z_J = 0} f(z)\Big\},$$

so that $D_k \geq B$ and $D_{n-1} = B$. The *standalone ratio* is $\rho = \min(\{1\} \cup \{a_j / c(e_j) : c(e_j) > 0\})$; the floor $\rho \geq r$ means $c(e_j) \leq a_j / r$ for every j. We study

$$\Gamma_k(w, r) = \sup\{(D_k - B)/L \ :\ \rho \geq r\},$$

the supremum over bounded demand laws and frozen offers. Write $s_k(w)$ for the sum of the $n - k$ largest shares.

Proposition 1 (Elementary ceilings)**.** *$\Gamma_k(w, 0) = s_k(w)$ for $1 \leq k \leq n - 2$, and $\Gamma_k(w, r) \leq (1 - r)s_k(w)$.*

Proof. On a face with free set F, $f(z) \leq a^\top z \leq a(F) \leq s_k(w)$ because $c \geq 0$; this bound is attained in the limit by two-state laws. For the second statement put $x = \sum_{j \in F} a_j(1 - z_j)$. Nonnegativity gives $f(z) - B \leq a(F) - B - x$; subadditivity and the caps give $c(\mathbf{1}_F) \leq c(z) + x/r$, hence $f(z) - B \leq (1/r - 1)x$. The weighted average of the two bounds with weights $1 - r$ and r is $(1 - r)(a(F) - B)$. □

The question is how much of the gap between $(1 - r)s_k(w)$ and the truth remains, and where.

3 Two tools

Lemma 2 (Laws from generators)**.** *Let $g_1, \dots, g_m \in \mathbb{R}^n$. The law that picks k uniformly and a sign $\pm$ uniformly and sets $Y = \mu \pm m g_k$ has $c(x) = \sum_k |g_k^\top x|$ for every x. After an affine change of units it takes values in $[0, 1]^n$, and its centre is an optimal grand offer with $L = c(\mathbf{1})$.*

Proof. $x^\top Y$ is symmetric about $x^\top \mu$, which is therefore an optimal offer, and $\mathbb{E}|x^\top(Y - \mu)| = \sum_k |g_k^\top x|$. Scaling all outputs, offers and costs by a common factor changes no ratio. □

Hence a value V is attained at depth k as soon as some generators satisfy the caps $c(e_j) \leq a_j / r$, stability $c(\mathbf{1}_S) \geq a(S)$ for every proper S (so $B = 0$), $c(\mathbf{1}) = 1$, and $a^\top z - c(z) = V$ at some $z \in [0, 1]^n$ with k zero coordinates. All constructions below are of this form.

Lemma 3 (Triangle certificates)**.** *Let z lie on a face with free set F, let $S \subseteq F$ and $0 \leq \nu \leq 1$. Then*

$$f(z) - B \leq a^\top z - \nu\, a(S) + \sum_{i \in S} \frac{a_i}{r} |z_i - \nu| + \sum_{i \in F \setminus S} \frac{a_i}{r} z_i.$$

Proof. $\nu \mathbf{1}_S = z - (z - \nu \mathbf{1}_S)$, so the triangle inequality and the caps give $\nu c(\mathbf{1}_S) \leq c(z) + \sum_{i \in F} (a_i / r) |z_i - \nu \mathbf{1}_S(i)|$, while $c(\mathbf{1}_S) \geq a(S) - B$. Since $\nu \leq 1$, the term $\nu B - B$ is nonpositive. □

By positive homogeneity a maximizing membership with positive excess can be scaled until its largest coordinate is one. With that coordinate and the order of the others fixed, every certificate with ν equal to a coordinate is linear in z, so the best bound from all of them is a linear program. It is valid for every demand law.

4 Equal shares

Let $a_j = 1/n$, write $M = a/r$ for the common cap, $m = n - k$ for the number of free producers, and for $p = 1, \dots, m-1$ put $q = m - p$ and

$$I_p = \Big[\max\Big(0, \tfrac{q-p}{q+p}\Big),\ \tfrac{q}{q+2p}\Big].$$

Theorem 4. *For equal shares, $1 \le k \le n-2$ and $0 < r \le 1$,*

$$\Gamma_k = \frac{(1-r)m}{n} \quad \text{if } r \in I_1 \cup \cdots \cup I_{m-1}, \qquad \Gamma_k = \frac{(m-1)(1-r^2)}{2nr} \quad \text{if } r \ge \frac{m-1}{m+1}.$$

Proof. The first value is the ceiling of Proposition 1. For the second, combine for each fractional coordinate the certificates of Lemma 3 with S containing or omitting it ($\nu = 1$): with the largest coordinate at one, $f(z) - B \le \sum_j h(z_j)$ where $h(t) = \min\{(1-t)(M-a), t(M+a)\}$, and also $f(z) - B \le a^\top z$. The minimum is concave and symmetric in the $m-1$ other coordinates, so a common value is optimal, and a one-variable computation gives the stated value for $r \ge (m-1)/(m+1)$.

Construction on I_p. Take p unit producers I, q fractional producers J, the excluded set K, $t = (1-r) - pr/q$, generators ae_l ($l \in K$) and $(Mt/p)\mathbf{1}_I - Me_j$ ($j \in J$), and $z = \mathbf{1}_I + t\mathbf{1}_J$. Every generator vanishes at z, so $c(z) = 0$ and $a^\top z = (1-r)m/n$. The unit caps read $qt \le p$, i.e. $r \ge (q-p)/(q+p)$; $c(\mathbf{1}) = ka + qM(1-t) = 1$. For a coalition with s unit, τ fractional and κ excluded members, $c(\mathbf{1}_S) - a|S| = \tau(M-a) + (q-2\tau)Mts/p - as$, bilinear in (s, τ), with corner values 0, $q(M-a)$, $qMt - ap \ge 0$ (i.e. $r \le q/(q+2p)$) and 0.

Construction for $r \ge (m-1)/(m+1)$. Take one unit producer u, the others J, $\delta = (M-a)/2$, $\beta = a - (m-1)\delta \ge 0$, generators βe_u, ae_l ($l \in K$), $\delta e_u - Me_j$ ($j \in J$), and $z = e_u + \frac{1-r}{2}\mathbf{1}_J$. Then $c(z) = \beta$, $c(e_u) = a$, $c(e_j) = M$, $c(\mathbf{1}) = 1$, and $c(\mathbf{1}_S) = a|S|$ when $u \in S$ and $c(\mathbf{1}_S) \ge a|S|$ otherwise. The excess at z is the stated value. □

The intervals I_p leave gaps when $p^2 + p < m$: the *window* between I_{p+1} and I_p is (r_L, r_R) with $r_L = (m-p-1)/(m+p+1)$ and $r_R = (m-2p)/m$. At depth one these are $(1/5, 1/3)$ for $n = 4$, $(1/3, 1/2)$ for $n = 5$, $(3/7, 3/5)$ for $n = 6$; their total length is between 0.13 and 0.20 for $4 \le n \le 30$ and decays only like $n^{-1/2}$.

Inside a window two further families compete. The family H_p keeps p unit producers and puts the others at $(1-r)/2$, with positive cost at the witness; it is valid for $r \ge (m-p)/(m+p)$ and has value

$$H_p(r) = \frac{(m-p)(1-r^2)}{2nr}.$$

The family V_p moves one producer from unit to fractional: with $q' = m - p - 1$,

$$t_1 = \frac{(p+1)(1-r)}{2(p+1) - m(1-r)}, \qquad t_2 = \frac{(p+1)r - 1 + t_1}{q'}, \qquad V_p(r) = \frac{1}{n}(p - 1 + (p+1)r + 2t_1).$$

Theorem 5. *Let $p^2 + p < m$, $r_L \le r \le r_R$ and $Q(r) = mr^2 + 2(p-1)r + 2 + 2p - m$. Then $\Gamma_k = V_p(r)$ if $Q(r) \ge 0$ and $\Gamma_k = H_{p+1}(r)$ if $Q(r) \le 0$. Equivalently $\Gamma_k = \max(V_p, H_{p+1})$, since $V_p - H_{p+1} = a(m+p+1)(r - r_L)\,Q(r)/(2rD)$ with $D = 2(p+1) - m(1-r) > 0$ on the window.*

Proof. Laws. For H_{p+1} write $p' = p+1$ for the number of unit producers and $q = m - p'$, and use the generators βe_i on the units with $\beta = ((p'+q)a - qM(1+r)/2)/p'$, ae_l on K, and $(\delta/p')\mathbf{1}_I - Me_j$ on the fractional producers, where $\delta = M(1-r)/2$; the witness is $\mathbf{1}_I + \frac{1-r}{2}\mathbf{1}_J$, and the stability slack is again bilinear with corners 0, $q(M-a)$, 0, 0. It is valid because $r \geq r_L = (m-p')/(m+p')$ makes $\beta \geq 0$. For V_p take one producer $*$ in transition and

$$\gamma = \frac{M(p+1)(1-r)}{2p}, \qquad \alpha = \frac{M((p-1)+(p+1)r)}{2q'},$$

$$\beta = \frac{M(m(1-r)-2p)}{2q'}, \qquad \delta = -\frac{M(2(p+1)-m(1-r))}{2},$$

with generators $(\alpha/p)\mathbf{1}_I + \beta e_* - Me_j$ for the other fractional producers, $\gamma\mathbf{1}_I + \delta e_*$, and ae_l on K, at $z = \mathbf{1}_I + t_1 e_* + t_2 \mathbf{1}_{J'}$. All generators vanish at z, every free producer sits exactly at its cap, $c(\mathbf{1}) = 1$, and for each of the two cases $* \in S$ and $* \notin S$ the stability slack is bilinear in the counts, with corner values 0, $q'(M-a)$, $p(M-a)$, $a((m-p)/r-(m+p))$ and $M-a$, $a(p/r+p)$, 0, 0, all nonnegative on the window. The witness is admissible: $t_1 \leq 1$ is $r \geq r_L$, and $t_2 \geq 0$ reduces to a quadratic inequality in $1-r$ that holds on the whole window.

Upper bounds. With Z_I the sum of the p largest coordinates, $u = z_{p+1}$ and Y the sum of the rest, four certificates suffice: nonnegative cost F_0; the $p+1$ largest coordinates with $\nu = 1$ (F_A); the whole face with $\nu = u$ (F_B) and with $\nu = 1$ ($F_{B'}$). If $Q(r) \geq 0$, the weights $\lambda_B = 1$, $\lambda_A = (M(m-2p) - ma)/(2M)$, $\lambda_0 = (M^2(2+2p-m) + 2aM(p-1) + ma^2)/(2aM)$ and the multiplier $M(2+2p-m) + am$ on $p - Z_I \geq 0$ cancel every variable; they are nonnegative because $r \leq r_R$, $Q(r) \geq 0$ and $r \geq r_L$, and the resulting constant is $V_p(r)$. If $Q(r) \leq 0$, the weights $\lambda_A = (M-a)/(2M)$ and a split of the remaining $(M+a)/(2M)$ between F_B and $F_{B'}$, with multipliers on $p - Z_I \geq 0$ and $1 - u \geq 0$, cancel every variable, and feasibility of nonnegative weights rearranges exactly to $Q(r) \leq 0$; the constant is $H_{p+1}(r)$. □

Corollary 6. *For equal shares, $\Gamma_k(w, r)$ is known in closed form for every n, k and r, and is attained by a finite symmetric law with at most n generators. For four producers at depth one,*

$$\Gamma_1 = \tfrac{3}{4}(1-r)\ (r \leq \tfrac{1}{5}), \quad \frac{2-r+3r^2}{2(1+3r)}\ (\tfrac{1}{5} \leq r \leq \tfrac{1}{3}), \quad \tfrac{3}{4}(1-r)\ (\tfrac{1}{3} \leq r \leq \tfrac{1}{2}), \quad \frac{1-r^2}{4r}\ (r \geq \tfrac{1}{2}).$$

At $r = 1/4$, for instance, four producers who each pay at least a quarter of their standalone cost can still face an audit that reports a saving of $31/56 \approx 55\%$ of the expected pooled cost while no actual coalition gains anything.

5 Arbitrary shares

Theorem 7. *For positive shares and $0 < r \leq 1$, $\Gamma_{n-2}(w, r) = \max_{i \neq j} \Phi_{ij}(r)$, where, with $x = a_i/a_j$,*

$$\Phi_{ij}(r) = \begin{cases} a_j(1-r^2)/(2r), & x \geq (1-r)/(2r), \\ (1-r)(a_i + a_j), & (1-r)/(1+r) \leq x \leq (1-r)/(2r), \\ a_i(1+r)(a_i+a_j)/(ra_j + (1+r)a_i), & x \leq (1-r)/(1+r). \end{cases}$$

Proof. At depth $n-2$ a face has two free producers; scale the witness to $z = e_i + te_j$. Lemma 3 with nonnegative cost, the singleton $\{i\}$, the pair $\{i, j\}$ and, through the triangle inequality $tc(e_j) \leq c(z) + c(e_i)$, the cap of i, bounds the excess by the lower envelope of $a_i + ta_j$, $ta_j(1+1/r)$, $(1-t)a_j(1/r-1)$ and $(1-t)a_i(1+1/r)$, whose maximum over t is Φ_{ij}. The three regimes are attained by the families C31, H and G of Theorem 8 with $|I| = |J| = 1$. □

Fix an excluded set K, a split of the free producers into units I and fractional producers J, and write $a(I)$, A_J for their share totals.

Theorem 8. *For positive shares and any depth k, $\Gamma_k(w,r)$ is at least the largest of the following values over all K and splits:*

(a) C31: $(1-r)(a(I)+A_J)$ *if* $\frac{1-r}{1+r}A_J \le a(I) \le \frac{1-r}{2r}A_J$*;*

(b) H: $A_J(1-r^2)/(2r)$ *if* $a(I) \ge \frac{1-r}{2r}A_J$*;*

(c) G: $a(I)+tA_J$ *with* $t = a(I)/(rA_J+(1+r)a(I))$*, if* $a(I) \le \frac{1-r}{1+r}A_J$*.*

Proof. In all three families the excluded producers carry generators $a_l e_l$ and the unit coefficients are proportional to the shares. (a) With $t = (1-r) - ra(I)/A_J$, one generator per $j \in J$ with entries $(a_j/r)ta_i/a(I)$ on the units and $-a_j/r$ at j; the witness $\mathbf{1}_I + t\mathbf{1}_J$ has zero cost, the unit caps are the lower condition, and with σ the unit share fraction of a coalition and $a(T)$ its fractional share, the stability slack is bilinear with corners 0, $A_J(1/r-1)$, $A_J t/r - a(I)$ and 0. (b) With $\Delta = A_J(1-r)/(2r)$, unit generators $(1-\Delta/a(I))a_i e_i$ and fractional generators with entries $(a_j/r)\frac{1-r}{2}a_i/a(I)$ on the units and $-a_j/r$ at j; the slack is $a(T)(1/r-1)(1-\sigma) \ge 0$. (c) With $c_j = a_j a(I)/(rtA_J)$, generators with entries $c_j t a_i/a(I)$ on the units and $-c_j$ at j; the unit caps are tight, $c(\mathbf{1}) = 1$, the witness $\mathbf{1}_I + t\mathbf{1}_J$ has zero cost, and the slack is bilinear with corners 0, $A_J(\lambda-1)$, $a(I)(1/r-1)$, 0, where $\lambda = a(I)/(rtA_J) \ge 1$. □

A fourth family, with one producer in transition as in V_p, has the value $a(I)(1+r) - a_*(1-r) + 2t_1 a_*$ with $t_1 = (1-r)(a(I)+a_*)/(ra(F)+a(I)+a_*-A_{J'})$; we have not yet written general validity conditions for it and check each candidate exactly.

Conjecture 9. *For positive shares, every depth k and $0 < r \le 1$, $\Gamma_k(w,r)$ equals the largest value of the four families over all excluded sets and splits.*

The lower bound is proved, so the conjecture concerns the upper bound. For every instance it can be tested with the certificate linear program of Section 3. In 1000 random instances with up to six producers, random depth and r, the best family was a valid law in exact arithmetic and its value equalled the certificate bound in all 1000 cases (C31 in 482, H in 467, the transitional family in 27, G in 24); the same held in 68 further instances. In those 68, the elementary ceiling $(1-r)s_k(w)$ was the answer in all 39 instances with $r \le 0.4$ and in none of the 15 with $r \ge 0.7$.

6 Strict stability and positive definite covariance

Proposition 10. *For $0 < r < 1$ the suprema in Theorems 4, 5, 7 and 8 are unchanged if the law must also have positive definite covariance and strictly negative excess for every proper coalition.*

Proof. Let X^s put mass 0.9 evenly on the atoms $\pm s(e_i - e_j)$ over all pairs and mass 0.1 on $\pm\zeta\mathbf{1}$. With $1.8s/n > 1$ every proper coalition has $c^s(\mathbf{1}_S) > a(S)$, and $c^s(\mathbf{1}) \le 1$ for small ζ; its covariance is positive definite. Mix a worst-case law at the ratio $r' = (1-\varepsilon)/\min_j(1/r - \varepsilon c^s(e_j)/a_j)$ with X^s, weights $1-\varepsilon$ and ε. Costs of centrally symmetric laws mix linearly, so the mixture is strictly stable, respects the caps at r, has grand cost at most one and covariance at least ε times that of X^s, and its excess at the witness tends to the supremum as $\varepsilon \to 0$, by continuity of the closed forms in r. □

7 Asymmetric penalties

In most balancing markets a shortfall is settled at a worse price than a surplus. With penalties λ_+ per unit of surplus and λ_- per unit of shortfall the loss is, up to a constant factor, the pinball loss $\ell_\tau(u) = \tau u^+ + (1-\tau)u^-$ of level $\tau = \lambda_+/(\lambda_+ + \lambda_-)$, and the coalition cost becomes $c(x) = \min_q \mathbb{E}\,\ell_\tau(x^\top Y - q)$. Write $\kappa = \max(\tau, 1-\tau)/\min(\tau, 1-\tau) \ge 1$ for the ratio of the two penalties and $\Gamma_k^\kappa(w,r)$ for the worst gap; $\kappa = 1$ is the symmetric case.

Lemma 11. *The cost c is sublinear and nonnegative, and $c(-x) \le \kappa\, c(x)$ for every x.*

Proof. $\ell_\tau(-u) = \ell_{1-\tau}(u) \le \kappa\, \ell_\tau(u)$ pointwise. □

Call (d, α, β) with $\alpha, \beta \ge 0$ and $\alpha/\beta \in [1/\kappa, \kappa]$ a κ-segment, with cost $\alpha(d^\top x)^+ + \beta(d^\top x)^-$.

Proposition 12. *Every finite sum of κ-segment costs is, up to a positive factor, the pinball cost of a finite demand law on $[0,1]^n$.*

Proof. Let $\tau \le 1/2$. For a segment, solve $\alpha = \tau A + (1-\tau)B$, $\beta = (1-\tau)A + \tau B$; then $A, B \ge 0$ exactly when $\alpha/\beta \in [1/\kappa, \kappa]$. A variable V equal to A/τ and $-B/\tau$ with probability τ each and to 0 otherwise has 0 as a τ- and a $(1-\tau)$-quantile, and $\mathbb{E}\,\ell_\tau(sV) = \alpha s^+ + \beta s^-$ for every real s. Mixing such components along the directions d keeps 0 optimal for each of them, so their costs add; an affine map puts the law in the box. □

Corollary 13. *For every $\kappa \ge 1$: (a) $\Gamma^\kappa_k(w, r) \le (1-r)s_k(w)$; (b) every explicit law of Sections 4–6 has a counterpart with the same normalized costs, so every lower bound proved for $\kappa = 1$ holds for every κ.*

Proof. (a) The proof of the ceiling uses only subadditivity along the nonnegative difference $\mathbf{1}_F - z$. (b) Symmetric generators are κ-segments with $\alpha = \beta$. □

In the certificates of Lemma 3 a coordinate that has to be lowered is a shortfall direction, so it is charged $\kappa a_i/r$ instead of a_i/r per unit. With this one change the last-depth and equal-share arguments go through.

Theorem 14. *For positive shares, $0 < r \le 1$ and $\kappa \ge 1$, $\Gamma^\kappa_{n-2}(w, r) = \max_{i \ne j} \Psi_{ij}(r)$, where, with $x = a_i/a_j$,*

$$\Psi_{ij}(r) = \begin{cases} a_j(1-r)(r+\kappa)/(r(1+\kappa)), & x \ge \kappa(1-r)/(r(1+\kappa)), \\ (1-r)(a_i + a_j), & (1-r)/(r+\kappa) \le x \le \kappa(1-r)/(r(1+\kappa)), \\ a_i(r+\kappa)(a_i+a_j)/(a_i(r+\kappa) + r a_j), & x \le (1-r)/(r+\kappa). \end{cases}$$

For $\kappa = 1$ this is Theorem 7.

Proof. With $z = e_i + te_j$, the certificates for the pair at $\nu = 1$ and $\nu = t$, for $\{i\}$ at $\nu = 1$, and nonnegative cost bound the excess by $(1-t)a_j(1-r)/r$, $(1-t)a_i(r+\kappa)/r$, $ta_j(r+\kappa)/r$ and $a_i + ta_j$; the maximum over t of their minimum is Ψ_{ij}. The other producers get the segments (e_k, a_k, a_k), and one or two segments on the pair attain the three regimes. In the first, the segment along $te_i - e_j$ has $(\alpha, \beta) = (\kappa a_j/r, a_j/r)$ and $t = (1-r)/(1+\kappa)$, and a symmetric segment of weight $a_i - \kappa t a_j/r$ lies along e_i. In the second, the segment along $te_i - e_j$ has $\beta = a_j/r$, $t = 1 - r - rx$ and α in an interval that is nonempty exactly in that regime. In the third, the segment along $e_i - e_j/t$ has $(\alpha, \beta) = (a_i/r, \kappa a_i/r)$. □

For equal shares put $m = n - k$, $H^\kappa_p(r) = (m-p)(1-r)(r+\kappa)/(nr(1+\kappa))$ and

$$I^\kappa_p = \Big[\max\Big(0, \tfrac{m-(1+\kappa)p}{m}\Big),\ \tfrac{\kappa(m-p)}{\kappa m + p}\Big],$$

which for $\kappa = 1$ are H_p and I_p.

Theorem 15. *For equal shares and $\kappa \ge 1$, $\Gamma^\kappa_k(r) \le \min\{(1-r)m/n,\ H^\kappa_1(r)\}$ for every r, with equality on every I^κ_p and for $r \ge \kappa(m-1)/(\kappa m + 1)$. The gap between I^κ_{p+1} and I^κ_p is nonempty if and only if $\kappa < (p+1)(m-p)/(pm)$. In particular, if $\kappa \ge 2(m-1)/m$, and so whenever a shortfall costs at least twice as much as a surplus,*

$$\Gamma^\kappa_k(r) = \min\Big\{(1-r)\frac{m}{n},\ \frac{(m-1)(1-r)(r+\kappa)}{nr(1+\kappa)}\Big\} \qquad (0 < r \le 1).$$

Proof. With $1 = z_1 \geq z_2 \geq \dots$ and $\sigma = z_2 + \dots + z_m$, the certificates for the face at $\nu = 1$ and for $\{1\}$ at $\nu = 1$ bound the excess by $(1-r)(m-1-\sigma)/(nr)$ and $\sigma(r+\kappa)/(nr)$; their minimum is at most H_1^κ, with equality at $\sigma = (1-r)(m-1)/(1+\kappa)$. The ceiling law of Theorem 4, with p units and the others at $t = 1 - mr/(m-p)$, becomes one segment along $(t/p)\mathbf{1}_I - e_j$ per fractional producer with $\beta = 1/(nr)$ and α in an interval that is nonempty exactly on I_p^κ. The law H_p becomes the same directions with $(\alpha, \beta) = (\kappa, 1)/(nr)$ and $t = (1-r)/(1+\kappa)$, plus a symmetric segment on each unit. In both, the stability slack is bilinear in the numbers of units and of fractional producers in a coalition, so the corners suffice. The window condition follows from $(m-(1+\kappa)p)(\kappa m + p + 1) - m\kappa(m-p-1) = (1+\kappa)\big(m(p+1-p\kappa) - p(p+1)\big)$. □

For arbitrary shares at intermediate depths the same segments, applied to two blocks of producers, give a lower bound. For an excluded set K and a split of $F = N \setminus K$ into units I and fractional producers J, give each $j \in J$ one segment along $(ta_i/a(I))_{i\in I} - e_j$ with coefficients proportional to a_j, as in the three regimes of Theorem 14. The cost of a coalition then depends only on $a(S\cap I)/a(I)$ and $a(S\cap J)$, and its stability slack is bilinear in these two numbers. Hence

$$\Gamma_k^\kappa(w, r) \;\geq\; \max_{K,\,I} \; \Psi\big(a(I), a(F \setminus I)\big), \tag{1}$$

with Ψ the function of Theorem 14 evaluated at the two share totals; for $\kappa = 1$ this is the best of the families C31, H and G of Theorem 8. In 600 random instances at intermediate depths ($n \leq 6$), the bound (1) equals the certificate bound in all 299 instances with $\kappa \geq 2$. For $1 < \kappa < 2$ it does so in 201 of 204; the three exceptions resemble the windows below, with differences of at most $1.8 \cdot 10^{-3}$.

Conjecture 16. *For $\kappa \geq 2$, every depth and all positive shares, equality holds in (1).*

For equal shares the conjecture is Theorem 15. Its content is that the transitional family of Conjecture 9 is no longer needed once a shortfall costs at least twice as much as a surplus.

For four producers at depth one and $r = 1/4$ the worst gap rises from $31/56$ with symmetric penalties to $9/16$ with $\kappa = 2$. Asymmetry never helps: every exact value above is nondecreasing in κ, and as $\kappa \to \infty$ the elementary ceiling becomes exact on $[0, (m-1)/m]$ for equal shares. Inside the gaps that remain for $1 < \kappa < 2(m-1)/m$, the certificate bound equals the relaxation over all sublinear costs with $c(-x) \leq \kappa c(x)$. The best κ-segment law we found stays up to $1.3 \cdot 10^{-3}$ below it, so these narrow windows remain open.

8 Discussion

The elementary ceiling $(1-r)s_k(w)$ is the right answer more often than one might expect: for equal shares on most of the range of r, and for generic shares whenever producers pay a small or moderate part of their standalone cost. Where it is not, the exact curves show how much an auditor may discount a fractional audit, and the windows show that the correction is not confined to large r when shares are symmetric. Even with a substantial standalone floor the audit can report savings of half the pooled cost while no coalition gains.

Three questions remain open. The upper bound in Conjecture 9; whether the relaxation over all seminorms, which the explicit laws met in every case we computed, always coincides with the worst case over demand laws; and the narrow windows that asymmetric penalties leave open in Section 7, together with Conjecture 16.

Reproducibility. Every construction, closed form and computational claim is checked by scripts that write their outcomes to JSON before any summary is generated; exact rational arithmetic is used for all constructions, and the algebra of every proof (identities, corner values, dual weights and the sign factorizations) is verified symbolically for general sizes.

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